

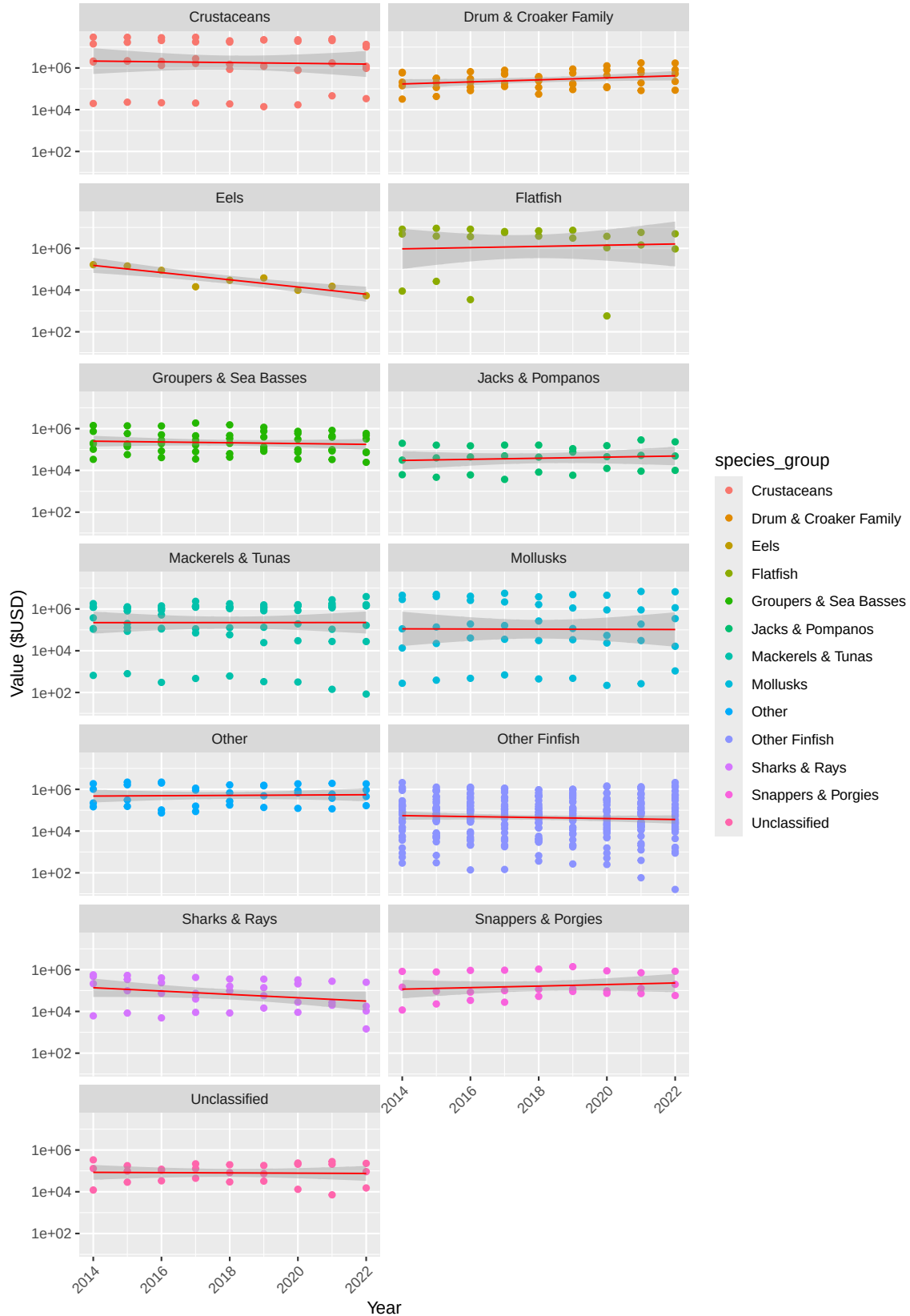
SchirmacherAyden_FinalProject

2025-04-01

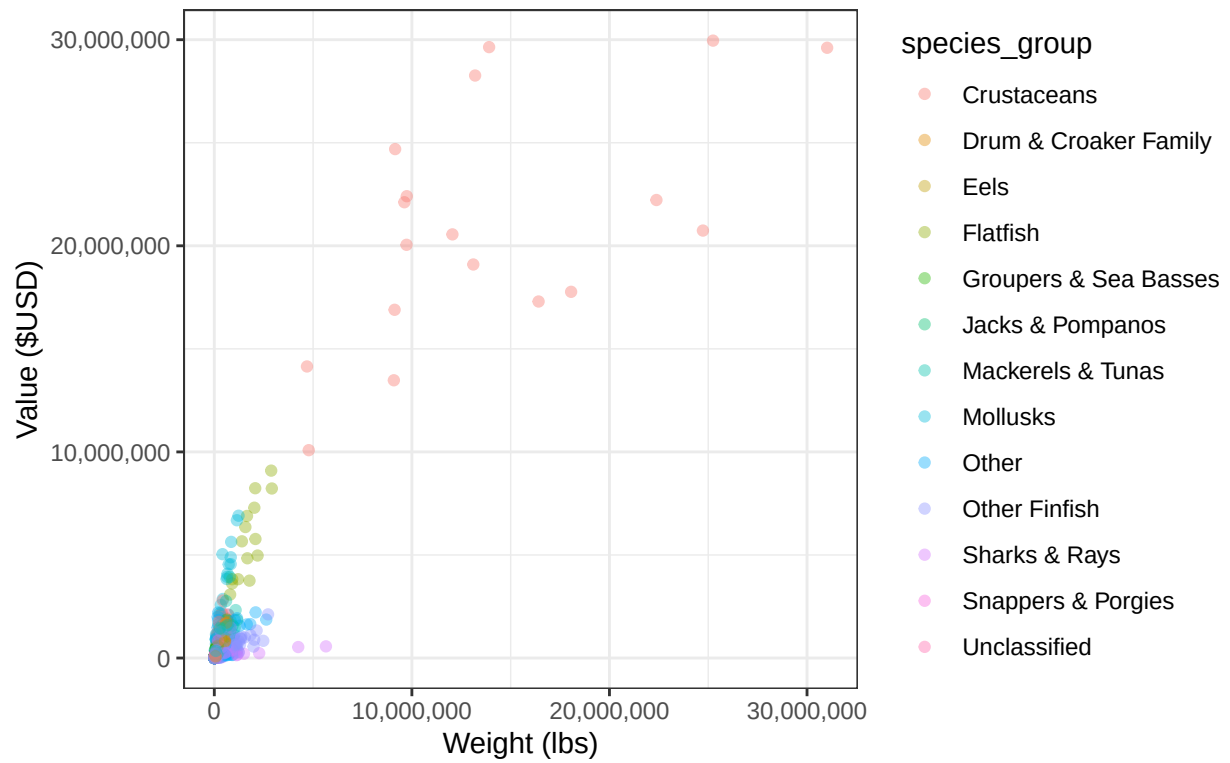
##Data Wrangling

##Exploratory Plots – Catch and Species Data

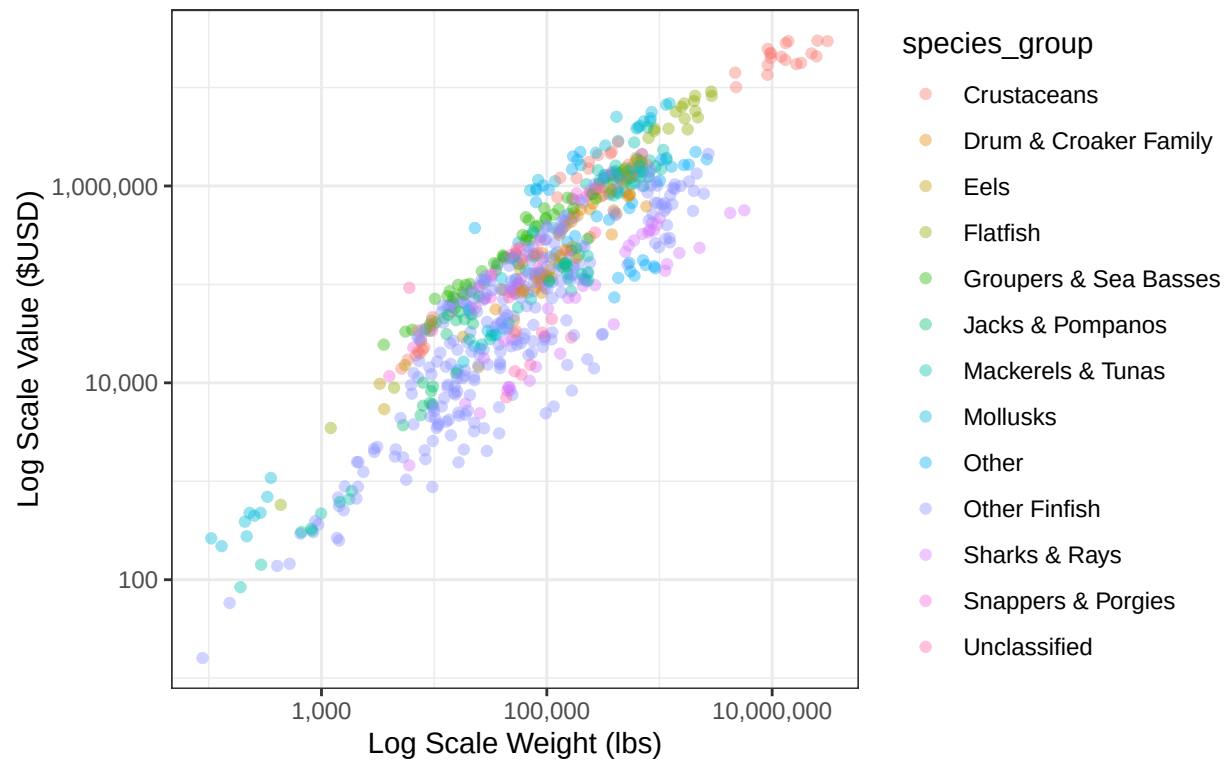
Final Value from Commercial Fishing
by Species Type in North Carolina, 2014–2022



Ratio of Fish Weight to Value from Commercial Fishing
by Species Type in North Carolina, 2014–2022

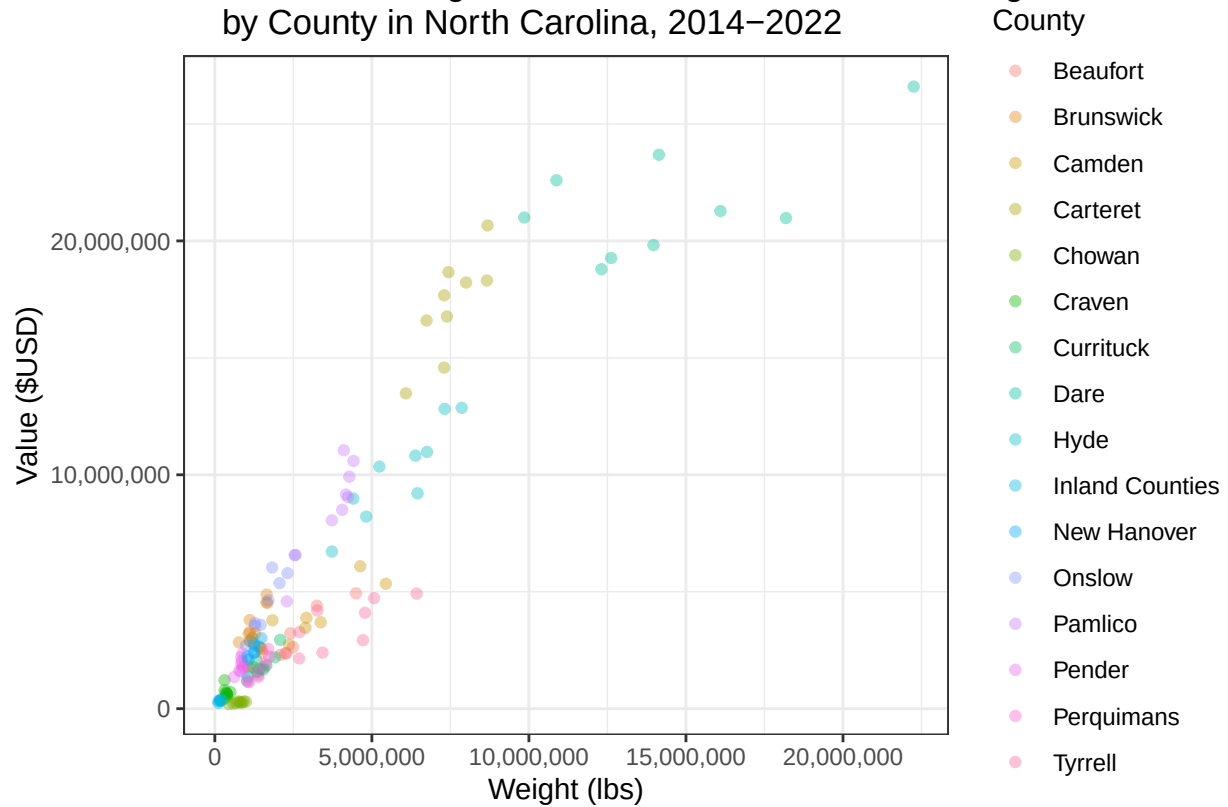


Logged Ratio of Fish Weight to Value from Commercial Fishing by Species Type in North Carolina, 2014–2022



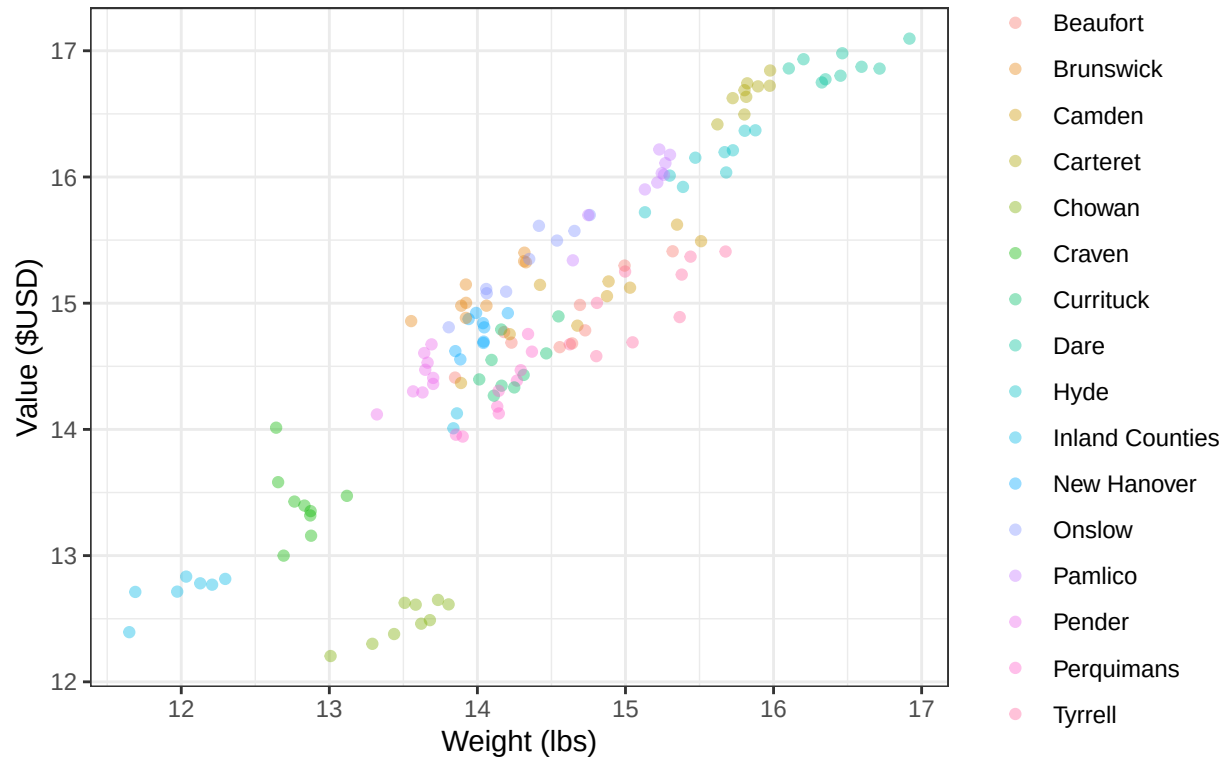
##Exploratory Plots – Spatial and County Data

Ratio of Fish Weight to Value from Commercial Fishing
by County in North Carolina, 2014–2022

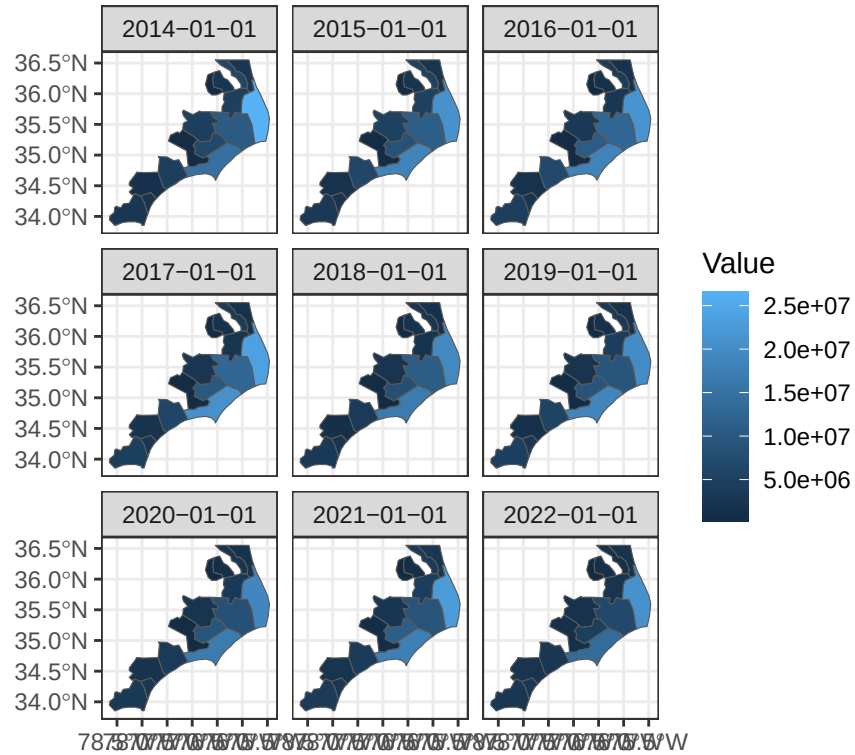


```
ggplot(fish_clean, aes(x = log(Pounds), y = log(Value), col=County))+
  geom_point(alpha = 0.40)+
  labs(
    title="Ratio of Fish Weight to Value from Commercial Fishing
    by County in North Carolina, 2014-2022",
    x="Weight (lbs)", y="Value ($USD)"+
    scale_y_continuous(labels=comma)+
  scale_x_continuous(labels=comma)+
  theme_bw()
```

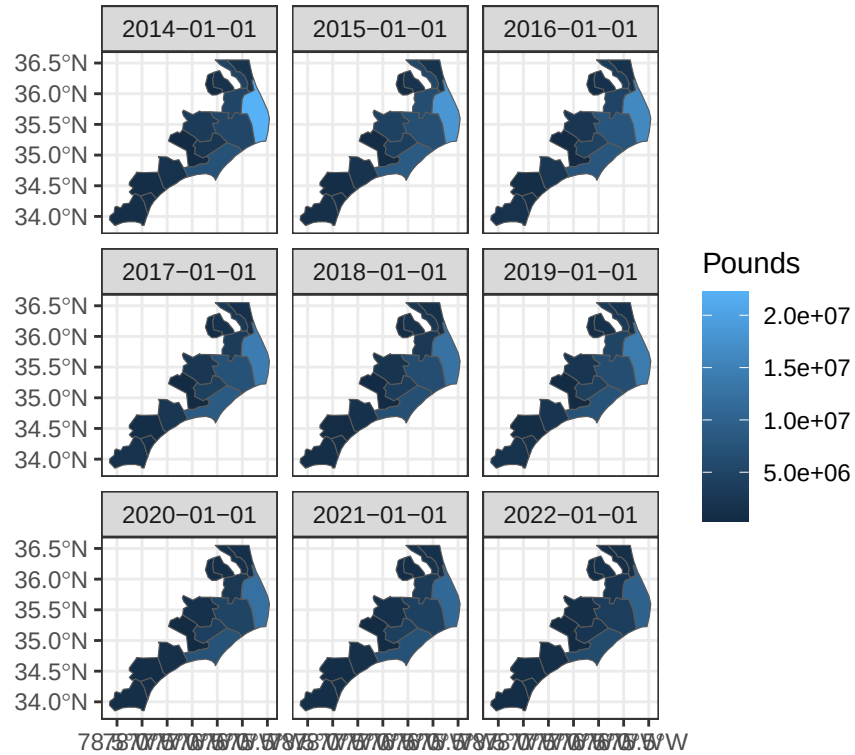
Ratio of Fish Weight to Value from Commercial Fishing
by County in North Carolina, 2014–2022



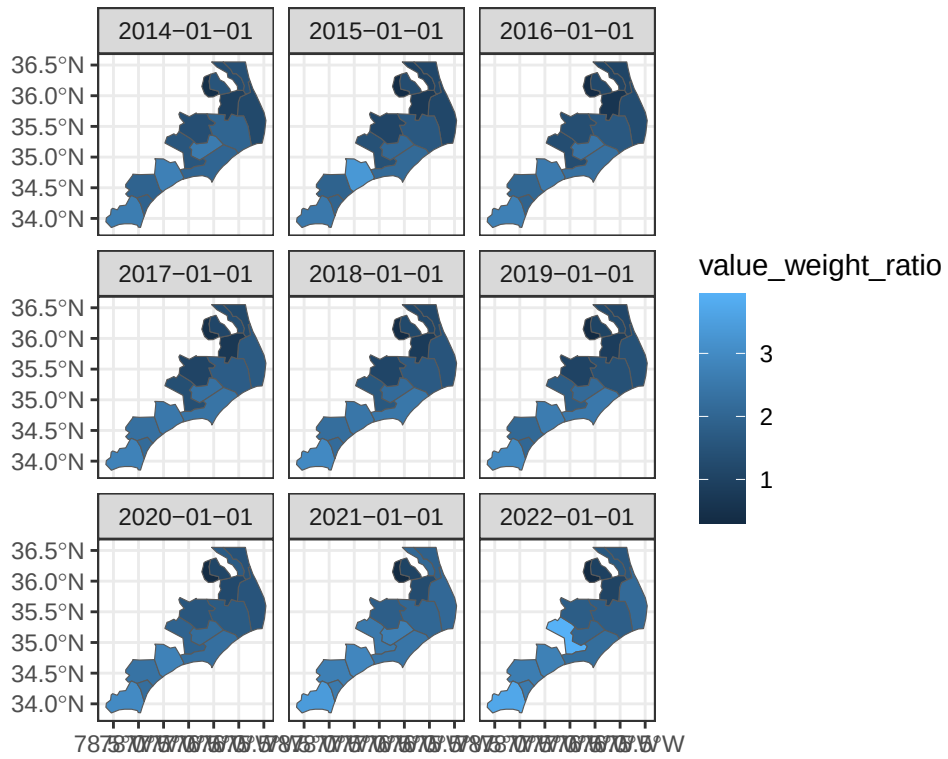
Value of Commercial Seafood Landings by County of Landing in North Carolina, 2014–2023



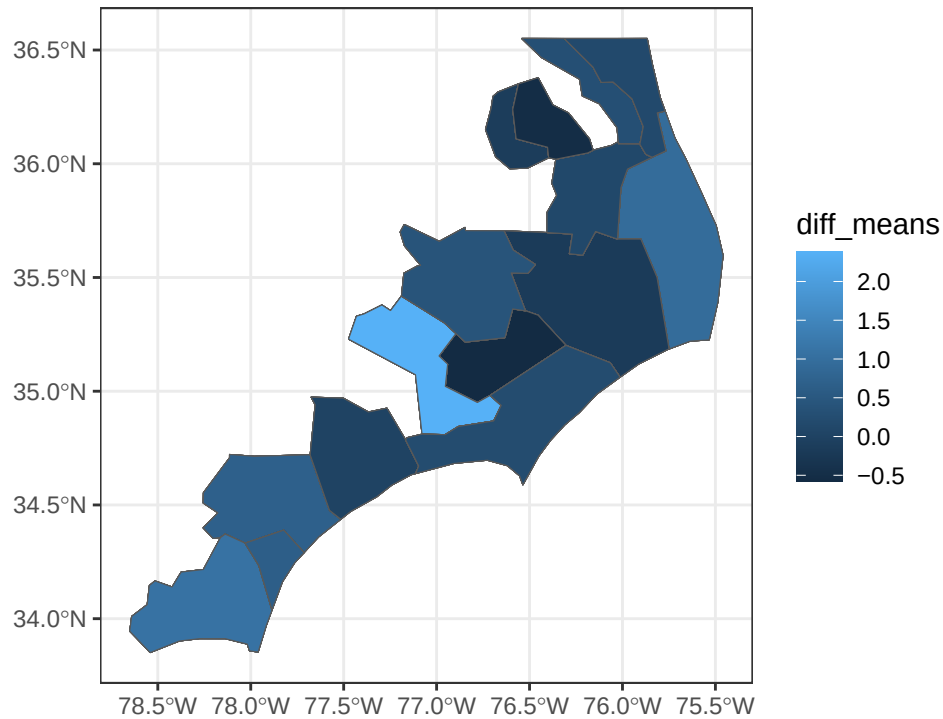
Weight of Commercial Seafood Landings by County of Landing in North Carolina, 2014–2023



Weight-Value Ratio of Commercial Seafood Landings by County of Landing in North Carolina, 2014-2023



Difference in Weight–Value Ratio of Commercial Seafood Landings Between 2014 and 2023 by County of Landing in North Carolina



Statistical Analysis – Catch and Species Data

Two sample t-test for value-to-weight ratio 2014 vs. 2022

```
## # A tibble: 12 x 2
##   species_group      p_value
##   <chr>             <dbl>
## 1 Crustaceans        0.541
## 2 Drum & Croaker Family 0.295
## 3 Flatfish          0.645
## 4 Groupers & Sea Basses 0.0143
## 5 Jacks & Pompanos    0.513
## 6 Mackerels & Tunas   0.358
## 7 Mollusks          0.414
## 8 Other              0.901
## 9 Other Finfish      0.219
## 10 Sharks & Rays     0.765
## 11 Snappers & Porgies 0.275
## 12 Unclassified      0.340
```

Linear Model Value across all years

```
##
## Call:
## lm(formula = FINALVALUE ~ Year, data = fish_all)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1173016 -1010676  -909332  -338357 28781431
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2502988.87 2364134.50   1.059   0.290
## Year          -82.74    134.69  -0.614   0.539
##
## Residual standard error: 3459000 on 739 degrees of freedom
## (960 observations deleted due to missingness)
## Multiple R-squared:  0.0005104, Adjusted R-squared:  -0.0008421
## F-statistic: 0.3774 on 1 and 739 DF, p-value: 0.5392
```

Linear Model Weight across all years

```
##
## Call:
## lm(formula = FINALWEIGHT ~ Year, data = fish_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1034081  -767992  -565652  -324540 30056943
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4197208.5 1905098.4   2.203   0.0279 *
## Year          -196.8    108.5  -1.813   0.0702 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2821000 on 757 degrees of freedom
## (942 observations deleted due to missingness)
## Multiple R-squared:  0.004325, Adjusted R-squared:  0.003009
## F-statistic: 3.288 on 1 and 757 DF, p-value: 0.07018
```

Linear Model RATIO across all years

```
##
## Call:
## lm(formula = log(FINALWEIGHT) ~ log(FINALVALUE), data = fish_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.45827 -0.66070 -0.06017  0.52568  2.98876
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.35806    0.16997   13.87  <2e-16 ***
## log(FINALVALUE) 0.76999    0.01428   53.91  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.9344 on 739 degrees of freedom
## (960 observations deleted due to missingness)
## Multiple R-squared: 0.7973, Adjusted R-squared: 0.797
## F-statistic: 2907 on 1 and 739 DF, p-value: < 2.2e-16
```

MLR ratio across all years

```
##
## Call:
## lm(formula = log(FINALVALUE) ~ log(FINALWEIGHT) + Year + species_group,
## data = fish_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9014 -0.4839  0.0520  0.4975  2.5638
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.222e+00  6.747e-01  -1.811  0.07059 .
## log(FINALWEIGHT)  1.037e+00  1.733e-02  59.859 < 2e-16 ***
## Year           9.849e-05  3.427e-05   2.874  0.00418 **
## species_groupDrum & Croaker Family -4.404e-01  1.865e-01  -2.362  0.01845 *
## species_groupEels -1.342e-01  3.269e-01  -0.411  0.68147
## species_groupFlatfish  1.222e-01  2.283e-01   0.535  0.59267
## species_groupGroupers & Sea Basses  5.836e-01  1.828e-01   3.192  0.00147 **
## species_groupJacks & Pompanos -5.931e-01  2.203e-01  -2.692  0.00727 **
## species_groupMackerels & Tunas -5.264e-01  1.687e-01  -3.121  0.00188 **
## species_groupMollusks  2.520e-01  1.919e-01   1.314  0.18942
## species_groupOther -6.290e-01  1.964e-01  -3.203  0.00142 **
## species_groupOther Finfish -1.013e+00  1.473e-01  -6.873 1.35e-11 ***
## species_groupSharks & Rays -2.249e+00  1.984e-01 -11.332 < 2e-16 ***
## species_groupSnappers & Porgies  1.827e-01  2.178e-01   0.839  0.40181
## species_groupUnclassified -8.081e-01  2.169e-01  -3.725  0.00021 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8769 on 726 degrees of freedom
## (960 observations deleted due to missingness)
## Multiple R-squared: 0.8696, Adjusted R-squared: 0.867
## F-statistic: 345.7 on 14 and 726 DF, p-value: < 2.2e-16
```

##Statistical Analysis – County and Spatial Data

Two sample t-test for value-to-weight ratio 2014 vs. 2022

Linear Model Value across all years

```
##
## Call:
## lm(formula = Value ~ Year, data = fish_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5741695 -3707750 -2414933  377310 20655071
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 12401573.9  9468113.9   1.310   0.192
## Year        -402.0      539.3  -0.745   0.457
##
## Residual standard error: 6103000 on 142 degrees of freedom
## Multiple R-squared:  0.003898, Adjusted R-squared:  -0.003117
## F-statistic: 0.5556 on 1 and 142 DF, p-value: 0.4573
```

Linear Model Weight across all years

```
##
## Call:
## lm(formula = Pounds ~ Year, data = fish_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3739897 -2043303 -1299356   882512 18293587
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 12938889.6  5726016.0   2.260   0.0254 *
## Year        -558.8      326.1  -1.713   0.0888 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3691000 on 142 degrees of freedom
## Multiple R-squared:  0.02025, Adjusted R-squared:  0.01335
## F-statistic: 2.935 on 1 and 142 DF, p-value: 0.08884
```

Linear Model RATIO across all years

```
##
## Call:
## lm(formula = log(Pounds) ~ log(Value), data = fish_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08064 -0.35346 -0.03665   0.27798   1.21326
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.41873    0.49101   4.926 2.3e-06 ***
## log(Value)   0.80654    0.03292  24.498 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4765 on 142 degrees of freedom
## Multiple R-squared:  0.8087, Adjusted R-squared:  0.8073
## F-statistic: 600.1 on 1 and 142 DF, p-value: < 2.2e-16
```

MLR ratio across all years

```
##
## Call:
## lm(formula = Value ~ Pounds + Year + County, data = fish_clean)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-3106454	-309839	-45526	355540	3489527

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.840e+06	1.825e+06	1.008	0.315215
Pounds	6.737e-01	7.928e-02	8.499	4.64e-14 ***
Year	-2.551e+01	9.717e+01	-0.263	0.793361
CountyBrunswick	1.438e+06	4.691e+05	3.065	0.002661 **
CountyCamden	3.572e+05	4.635e+05	0.771	0.442346
CountyCarteret	1.076e+07	6.176e+05	17.429	< 2e-16 ***
CountyChowan	-1.640e+06	4.780e+05	-3.432	0.000812 ***
CountyCraven	-9.478e+05	4.870e+05	-1.946	0.053836 .
CountyCurrituck	-3.810e+05	4.657e+05	-0.818	0.414775
CountyDare	1.041e+07	1.068e+06	9.750	< 2e-16 ***
CountyHyde	4.749e+06	5.406e+05	8.785	9.61e-15 ***
CountyInland Counties	-1.087e+06	4.873e+05	-2.231	0.027466 *
CountyNew Hanover	4.059e+05	4.699e+05	0.864	0.389356
CountyOnslow	2.103e+06	4.640e+05	4.532	1.34e-05 ***
CountyPamlico	4.682e+06	4.751e+05	9.856	< 2e-16 ***
CountyPender	-9.860e+04	4.767e+05	-0.207	0.836465
CountyPerquimans	-6.687e+05	4.669e+05	-1.432	0.154613
CountyTyrrell	-5.848e+05	4.783e+05	-1.223	0.223772

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 978900 on 126 degrees of freedom
## Multiple R-squared:  0.9773, Adjusted R-squared:  0.9742
## F-statistic: 318.6 on 17 and 126 DF, p-value: < 2.2e-16
```